

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0605

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

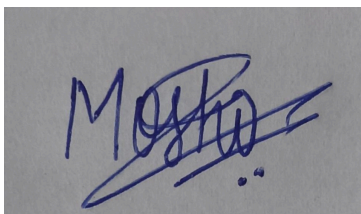
Annexure A

CONCEPT MAPPING

Code of Conduct:

*I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

A handwritten signature in blue ink, appearing to read 'Moshika', with a stylized flourish extending from the end.

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response.

(5 Marks)

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

(5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

6. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only. (5 Marks)

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

7. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks)

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

8. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the **specific (S)** and **general (G)** boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

9. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the **version space (S and G boundaries)** and show the updates after each example. (5 Marks)

Dataset:

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. (5 Marks)

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing,

feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

12. Design a machine learning model for spam email classification using labeled data. (5 Marks)

- (a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
- (b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.
- (c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks)

Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept *EnjoySport*.

- (a) Explain the initialization of S (specific boundary) and G (general boundary).
- (b) Show how S and G change with each training example.
- (c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)

- (a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).
- (b) Explain feature scaling and handling missing data.
- (c) Describe how overfitting can be detected and avoided.

15. Explain the concept of clustering in unsupervised learning. (5 Marks)

- (a) Describe the working of the K-Means algorithm with a step-by-step example.
- (b) Explain how the value of K is chosen using the Elbow Method.
- (c) Compare K-Means with Hierarchical Clustering.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. MACHINE LEARNING SYSTEM – PIPELINE WITH SUPERVISED & UNSUPERVISED LEARNING

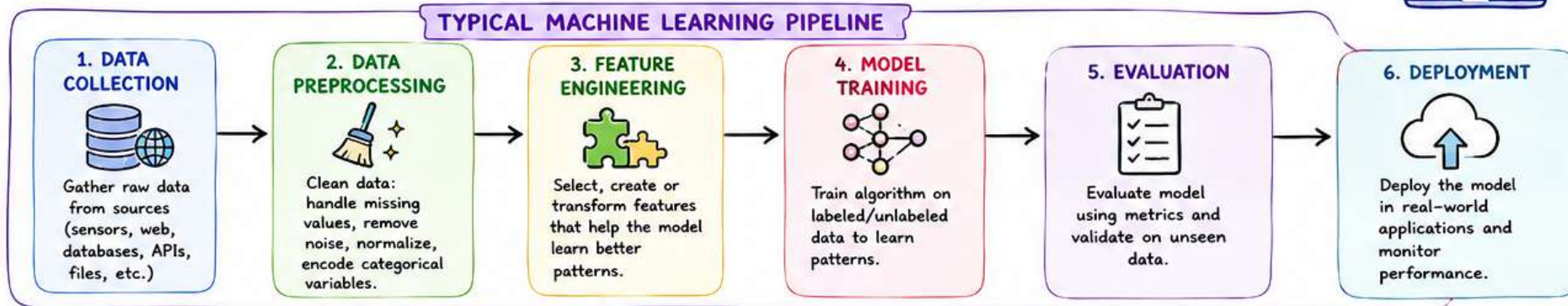


GOAL

Build systems that learn patterns from data and make useful predictions or discover hidden structure.



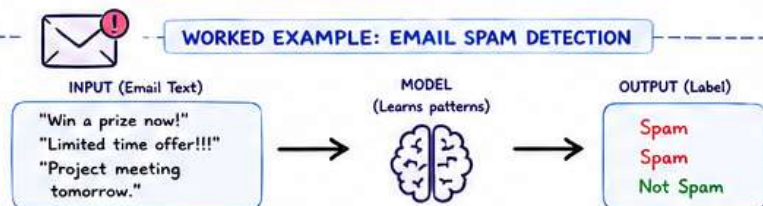
TYPICAL MACHINE LEARNING PIPELINE



SUPERVISED LEARNING (Labeled Data)

Learn a mapping from inputs (X) to known outputs (Y). Used for prediction.

WORKED EXAMPLE: EMAIL SPAM DETECTION



- **Data:** Emails labeled as Spam or Not Spam
- **Task Type:** Classification (binary)
- **What model learns:** Words, phrases, structure associated with Spam
- **Possible Algorithms:** Logistic Regression, Naïve Bayes, SVM, Decision Tree, Random Forest, Neural Networks

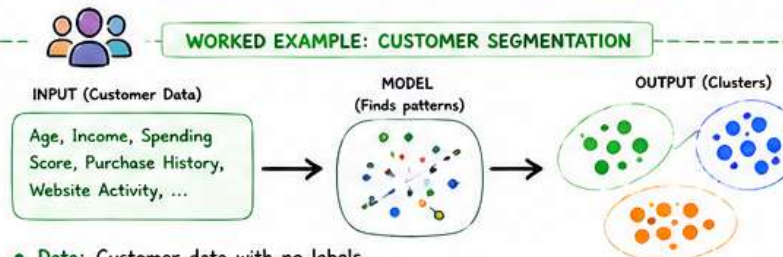
KEY POINT

Supervised uses labeled data to predict outputs.
Unsupervised uses unlabeled data to discover patterns/structure.

UNSUPERVISED LEARNING (Unlabeled Data)

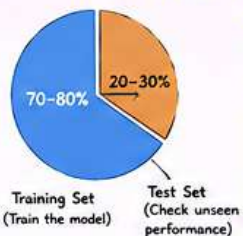
Find hidden patterns or structure in data without known outputs. Used for exploration.

WORKED EXAMPLE: CUSTOMER SEGMENTATION



- **Data:** Customer data with no labels
- **Task Type:** Clustering
- **What model finds:** Groups of similar customers
- **Possible Algorithms:** K-Means, Hierarchical Clustering, DBSCAN, Gaussian Mixture Models (MMM), Apriori (Association Rules)

TRAIN / TEST SPLIT & EVALUATION METRICS



- Split data into training (70-80%) and test (20-30%) sets.
- Use cross-validation (e.g., k-fold) for robust estimate.

For Classification:

Accuracy, Precision, Recall, F1-score, ROC-AUC

For Regression:

MAE, MSE, RMSE, R^2 Score

For Clustering:

Silhouette Score, Inertia (WCSS)



COMMON ALGORITHMS

Supervised

Classification: Logistic Regression, SVM, Decision Tree, Random Forest, Naïve Bayes, KNN, Neural Networks

Regression: Linear Regression, Ridge, Lasso, SVR, Decision Tree Regressor, Random Forest Regressor

Unsupervised

Clustering: K-Means, Hierarchical Clustering, DBSCAN, GMM

Association Rules: Apriori, FP-Growth

Dimensionality Reduction: PCA, t-SNE, UMAP



KEY TAKEAWAYS

- ✓ ML pipeline: Data → Preprocess → Features → Train → Evaluate → Deploy.
- ✓ Supervised predicts known outputs (labels).
- ✓ Unsupervised discovers hidden patterns/structure.
- ✓ Right data, features and evaluation lead to better models.
- ✓ Always monitor deployed models for data drift and performance.

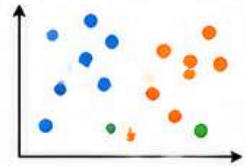


Q2. CLUSTERING IN UNSUPERVISED LEARNING 🔍

Discover natural groups (clusters) in data **without** using labels.

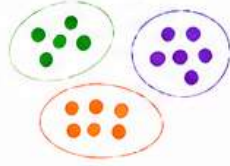
CLUSTERING OVERVIEW

Input: Unlabeled data (only features)



Clustering Algorithm

Output: Groups/Clusters



- ★ **Goal:** Group similar points together so that points in the same cluster are very similar and points in different clusters are dissimilar.
- ★ **Uses:** Pattern discovery, segmentation, recommendations, anomaly detection, image compression, bioinformatics, market analysis.
- ★ **No target variable is used.**

CLUSTERING QUALITY MEASURES



- **Silhouette Score:** Measures how similar a point is to its own cluster compared to other clusters. Range [-1, 1] (higher is better).



- **Inertia / WCSS:** Within-Cluster Sum of Squares. Lower values indicate tighter clusters.



- **Davies-Bouldin Index:** Average similarity between clusters. Lower is better.

(a) K-MEANS ALGORITHM

Goal: Partition data into K clusters by minimizing the within-cluster sum of squares (WCSS).

STEP-BY-STEP WORKING

- 1 Choose K**
Decide the number of clusters (K). 
- 2 Initialize Centroids**
Randomly select K points as initial centroids (μ). 
- 3 Assign Points**
Assign each data point to the nearest centroid (based on distance, usually Euclidean). 
- 4 Update Centroids**
Recalculate each centroid as the mean of all points assigned to it. 
- 5 Repeat**
Go back to Step 3 and repeat until assignments no longer change (converged). 

K-MEANS OBJECTIVE FUNCTION

Minimize Within-Cluster Sum of Squares (WCSS):

$$J = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2$$

where C_i = points in cluster i

μ_i = centroid of cluster i

$\|x - \mu_i\|^2$ = squared Euclidean distance.

NOTES

- ✓ Sensitive to initial centroids.
- ✓ Works well for spherical, similar-sized clusters.
- ✓ Requires K in advance.
- ✓ Scales well to large datasets.

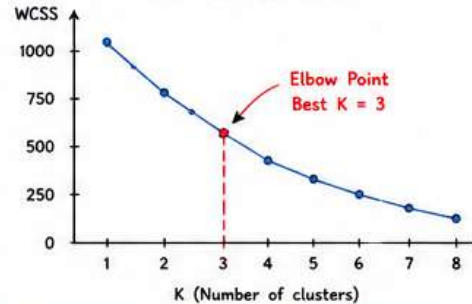
(b) ELBOW METHOD (CHOOSING K)

Idea: Run K-Means for different values of K and plot WCSS. Choose K at the "elbow" point where the rate of decrease sharply changes.

STEPS

- 1 Run K-Means for $K = 1, 2, 3, \dots, n$.
- 2 For each K, compute WCSS.
- 3 Plot K vs WCSS.
- 4 Choose K at the elbow point.

EXAMPLE ELBOW PLOT



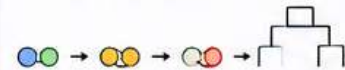
Interpretation: After $K = 3$, the improvement in WCSS becomes small. So, 3 clusters is a good choice.

(c) K-MEANS vs HIERARCHICAL CLUSTERING

Aspect	K-Means (Partitioning)	Hierarchical Clustering (Hierarchical)
Approach	Partition-based (flat)	Tree-based (nested)
Need K in advance?	Yes	No (choose by cutting dendrogram)
Output	K non-overlapping clusters	Dendrogram (full hierarchy)
Distance Metric	Usually Euclidean	Any (Euclidean, Manhattan, Cosine, etc.)
Scalability	Efficient for large datasets	Computationally expensive ($O(n^2)$ memory)
Sensitivity	Sensitive to initial centroids and outliers	More stable, deterministic
Cluster Shape	Works best for spherical, similar-sized clusters	Can capture arbitrary shapes
Update	Recomputed iteratively	Agglomerative (bottom-up) or Divisive (top-down)

Hierarchical - Agglomerative (Bottom-Up)

- Start with each point as its own cluster.
- Repeatedly merge the closest clusters until one cluster remains.



Hierarchical - Divisive (Top-Down)

- Start with all points in one cluster.
- Recursively split the cluster into smaller clusters.



TYPES OF CLUSTERING

- 1. Partitioning**
(K-Means, K-Medoids)
Fast, scalable, needs K.



- 2. Hierarchical**
(Agglomerative/Divisive)
Builds a tree of clusters.



- 3. Density-Based**
(DBSCAN)
Finds dense regions, handles noise/outliers.



- 4. Model-Based**
(GMM)
Assumes data comes from probability distributions.



APPLICATIONS



- Customer Segmentation
- Recommender Systems
- Image Segmentation
- Document Clustering



- Anomaly Detection
- Bioinformatics (genes)
- Social Network Analysis
- Market Basket Analysis

KEY TAKEAWAYS

- ✓ Clustering finds hidden structure in unlabeled data.
- ✓ K-Means is simple and fast but needs K and good initialization.
- ✓ Elbow Method helps choose the best K using WCSS.
- ✓ Hierarchical Clustering gives a hierarchy (dendrogram) and no need for K.
- ✓ Evaluate clusters using Silhouette Score or Inertia.





Q3 – FIND-S ALGORITHM : HYPOTHESIS UPDATES (PLAYTENNIS)

Find the most specific hypothesis consistent with **positive examples only** (ignore negative examples).



Goal: Learn a concept that predicts "Yes"

PLAYTENNIS DATASET

Ex	Outlook	Temp.	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No



IMPORTANT RULE

- Find-S uses **ONLY** the positive examples (Yes)
- Negative examples are completely **IGNORED**.

REPRESENTATION

Hypothesis h is a 4-tuple:

$h = \langle \text{Outlook, Temp., Humidity, Wind} \rangle$

SYMBOLS

- $\emptyset$ = most specific value (no information)
- $?$ = any value (most general for that attribute)
- Specific $\rightarrow$ General (order of generalization):
 - $\emptyset \rightarrow$ specific value $\rightarrow ?$

INTUITION

Start very specific. When a positive example doesn't match an attribute, generalize that attribute just enough (move toward "?" to cover it.



STEP-BY-STEP HYPOTHESIS UPDATES



Step	Example (Ex)	Play Tennis	Example (Outlook, Temp., Humidity, Wind)	Hypothesis h $\langle \text{Outlook, Temp, Humidity, Wind} \rangle$	What Happened?
Start	-	-	-	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$	Initialize to most specific hypothesis
1	1	No	$\langle \text{Sunny, Hot, High, Weak} \rangle$	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$	Negative example $\rightarrow$ IGNORED
2	2	No	$\langle \text{Sunny, Hot, High, Strong} \rangle$	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$	Negative example $\rightarrow$ IGNORED
3	3	Yes	$\langle \text{Overcast, Hot, High, Weak} \rangle$	$\langle \text{Overcast, Hot, High, Weak} \rangle$	1st positive $\rightarrow$ take exact values
4	4	Yes	$\langle \text{Rainy, Mild, High, Weak} \rangle$	$\langle ?, ?, \text{High, Weak} \rangle$	Outlook & Temp. differ $\rightarrow$ generalize to "?"
5	5	Yes	$\langle \text{Rainy, Cool, Normal, Weak} \rangle$	$\langle ?, ?, ?, \text{Weak} \rangle$	Humidity differs (High $\neq$ Normal) $\rightarrow$ "?"
6	6	No	$\langle \text{Rainy, Cool, Normal, Strong} \rangle$	$\langle ?, ?, ?, \text{Weak} \rangle$	Negative example $\rightarrow$ IGNORED
7	7	Yes	$\langle \text{Overcast, Mild, Normal, Strong} \rangle$	$\langle ?, ?, ?, ? \rangle$	Wind differs (Weak $\neq$ Strong) $\rightarrow$ "?"
8	8	No	$\langle \text{Sunny, Mild, High, Weak} \rangle$	$\langle ?, ?, ?, ? \rangle$	Negative example $\rightarrow$ IGNORED

Positive examples used: Ex. 3, 4, 5, 7

FINAL MAXIMALLY SPECIFIC HYPOTHESIS

$h = \langle ?, ?, ?, ? \rangle$

INTERPRETATION



All attributes are generalized to "?". Based only on the four positive examples, no single attribute value is reliable to predict "Yes". More data (especially informative positives and negatives) is needed to specialize further.



FIND-S ALGORITHM (PSEUDOCODE)



- Initialize h to the most specific hypothesis:
 $h \leftarrow \langle \emptyset, \emptyset, \dots, \emptyset \rangle$
- For each training example x in the data:
 - If x is positive (Yes):
For each attribute a_i in h
 - If h_i satisfies x , leave it unchanged;
 - Else replace h_i with the next more general value satisfying x .
 - If x is negative (No): **IGNORE** it.
- Output h as the final hypothesis.



ADVANTAGES

- Simple and easy to understand
- Guaranteed to find a hypothesis consistent with all positive examples (if one exists)
- Low computational cost. (single pass)
- Works well when target concept is simple and noise is minimal



LIMITATIONS

- Ignores negative examples completely
- Cannot detect inconsistent data
- No way to know if final hypothesis is the only consistent one
- Sensitive to noise – a single wrong positive can mislead the hypothesis



KEY TAKEAWAY

Find-S moves from most specific ($\emptyset$) to more general ($\rightarrow ?$) using only positive examples, producing the most specific hypothesis that covers all positives.

Final h for this dataset: $\langle ?, ?, ?, ? \rangle$



Q4. CANDIDATE ELIMINATION ALGORITHM - VERSION SPACE (S & G BOUNDARIES)



Goal: Find the version space by computing the Specific (S) and General (G) boundaries using both positive and negative examples.

ENJOY-SPORT DATASET

Ex	Weather	Temp.	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

☑ Yes = Positive Example ☒ No = Negative Example

KEY IDEA

- ☀ **S (Specific Boundary):** Most specific hypothesis consistent with all positive examples seen so far.
- ☀ **G (General Boundary):** Set of most general hypotheses consistent with all negative examples seen so far and not inconsistent with S.

INITIALIZATION

At the beginning (before seeing any examples):

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$ (most specific hypothesis)

$G_0 = \langle ?, ?, ?, ? \rangle$ (most general hypothesis)

Where: $\emptyset$ = most specific value (no information)
? = any value (most general)

HOW S & G ARE UPDATED

- ☑ If example is **POSITIVE (Yes)**:
Generalize S to cover the example (make it more general).
Remove from G any hypothesis that is inconsistent with it.
- ☒ If example is **NEGATIVE (No)**:
Specialize G to remove the example (make it more specific).
Remove from S any hypothesis that is inconsistent with it.

STEP-BY-STEP S & G BOUNDARY UPDATES

After Example	Example (Weather, Temp., Humidity, Wind, Enjoy?)	S (Specific Boundary)	G (General Boundary)	What Happened?
0 (Start)	-	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$	$\langle ?, ?, ?, ? \rangle$	Initialize S to most specific and G to most general.
1 (Yes)	(Sunny, Warm, Normal, Weak)	(Sunny, Warm, Normal, Weak)	$\langle ?, ?, ?, ? \rangle$	Positive → Generalize S to match the example.
2 (Yes)	(Sunny, Warm, High, Strong)	(Sunny, Warm, ?, ?)	$\langle ?, ?, ?, ? \rangle$	Positive → Humidity and Wind differ → generalize to "?".
3 (No)	(Rainy, Cold, High, Weak)	(Sunny, Warm, ?, ?)	$\{ \langle \text{Sunny}, ?, ?, ? \rangle, \langle ?, \text{Warm}, ?, ? \rangle \}$	Negative → specialize G to exclude this example while keeping it consistent with S.
4 (Yes)	(Sunny, Cold, Normal, Weak)	(Sunny, ?, ?, ?)	$\{ \langle \text{Sunny}, ?, ?, ? \rangle \}$	Positive → Temp differs (Warm vs Cold) → generalize Temp to "?". Remove any G that is inconsistent with S.
5 (No)	(Rainy, Warm, Normal, Strong)	(Sunny, ?, ?, ?)	$\{ \langle \text{Sunny}, ?, ?, ? \rangle \}$	Negative → already rejected by G (Weather ≠ Sunny). So, no change.

CONVERGED VERSION SPACE

$S = G = \{ \langle \text{Sunny}, ?, ?, ? \rangle \}$

Interpretation: The target concept is "Sunny weather", other attributes can be anything.

FINAL RESULT

$S = G = \{ \langle \text{Sunny}, ?, ?, ? \rangle \}$

The algorithm converges to a single hypothesis.

KEY INSIGHTS

- ★ When $S = G$, the exact target concept is found.
- ★ Candidate Elimination uses both positive and negative examples.
- ★ It maintains the complete version space, not just one hypothesis (unlike Find-S).

CANDIDATE ELIMINATION ALGORITHM (PSEUDOCODE)

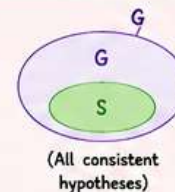
- Initialize: $S \leftarrow \langle \emptyset, \emptyset, \dots, \emptyset \rangle$ and $G \leftarrow \langle ?, ?, \dots, ? \rangle$
- For each training example d :
 - If d is **positive (Yes)**:
 - Remove from G any hypothesis inconsistent with d .
 - For each s in S inconsistent with d , replace s with all minimal generalizations that are consistent with d and more specific than some member of G.
 - If d is **negative (No)**:
 - Remove from S any hypothesis inconsistent with d .
 - For each g in G inconsistent with d , replace g with all minimal specializations that are consistent with d and more general than some member of S.
- Output the final S and G (version space).

SYMBOLS & MEANING

- $\emptyset$: most specific value (no info)
- ? : any value (most general)
- S : specific boundary (single hypothesis)
- G : general boundary (set of hypotheses)

VERSION SPACE

All hypotheses that lie between S and G (inclusive) are consistent with the training data.
As more examples are processed, the version space shrinks.



ADVANTAGES

- Uses both positive & negative examples.
- Finds the complete version space.
- Detects inconsistent training data.
- More powerful than Find-S.



LIMITATIONS

- Version space can grow exponentially.
- Assumes noise-free, correctly labeled data.
- Not suitable for disjunctive target concepts.